

Final Report

Project 2 - Event Extractor from Conference Emails

Team Members:

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Problem, Background, and Context:

People receive many conference emails in their inbox, and they can pile up very fast. Unfortunately, we have to manually go through each email in order to find out its respective conference event details and this gets inconvenient since there are a lot of conference emails. To fix that, our final project is to create an event extractor that can go through raw text, email files, or websites and retrieve proper accurate information and display them so users don't need to read through anything.

Data Sources, Tools, and Services:

For our website portion of the project, we are using React + Vite as our frontend, and Python for our backend. For the model making part, we use Python with our main key packages being spaCy, pandas, BeautifulSoup, sklearn, re, transformers, rapidfuzz, and duckdb. For website deployment, we use Render to deploy our backend, and Vercel for our frontend. Our data source that we use for testing and training data comes from Professor Foaad's excel sheet of previous students' conference email extractions. We also relied on using <https://conferenceindex.org/conferences> for testing.

Design of the System / Our Approach:

Our overall approach was to first clean out the data in the excel sheet reference (<https://docs.google.com/spreadsheets/d/1MbgeCI54iKuX0rab3ZtigPscGjD2lwOZGquG11sxo5A/edit?gid=1204578990#gid=1204578990>). After exporting the raw CSV files, we made a convertCSV.ipynb that took specific columns we needed from each CSV file and joined them all into one clean and organized CSV file (since not all raw files had the same data structure and naming). We used pandas, and email for this.

We also tagged two additional fields manually - published date and abstract date, and combined them with the existing CSV file. After that we focused on creating our training and testing data from our cleaned up CSV file. We used sklearn's train_test_split function to do so.

Then we started working on building our own models. We created a total of three models (the old models are in the OLD_WORK folder of our repository). We used Named Entity Recognition (NER) as our main technique in our model to search and extract our event information. We also used pattern date matching to help us search for date information easier. Then we trained our models with the train CSV file we created earlier.

Our final model's preprocess required getting the BIO tags of the human parsed tags for each email. Our process for doing this was to first feed each email through a pretrained roBERTa model. This is a bidirectional transformer that is better able to pick up context than our other efforts. After feeding the email in, each recognized entity was paired with the human parsed field by fuzzy matching each one and selecting the one with the best result. This allowed us to then convert each match into the appropriate BIO tag.

After creating the BIO tags, a base roBERTa model was fine tuned and retrained on the new labels. This involved feeding in 640 randomly sampled emails from our test train split.

To test our models, we built an evaluator to determine accuracy, precision, recall, and F1 scores for all of our models. The biggest issue when creating an evaluator is that the human parsed data often did not appear in the text. Only 56% of all the human parsed information actually appears in the text. Oftentimes the date will be listed in a format such as July 24-27, 2024, where the human parsed date is listed as July 24, 2024. To solve this problem our evaluator used fuzzy matching where a 30% fuzzy match was considered a match.

Finally, we worked on creating a website to display all the data. In the frontend, we display a table of rows which are extraction results from the json file exported after we run our model on the testing data. The website also displays all the scores and has two different input boxes where users can either paste raw text or submit a website link to extract event information from. After submitting input, a new row would be generated at the top of the table with extracted information.

One last thing we made sure to do is also make it runnable on the terminal itself. We created a terminal_parse.py where users can run the file and add an argument of either a .eml, .txt, or a website link, and get the extracted data.

Images:

Email Dashboard

Paste text to process

Process Text

Enter link to process

Process Link

Model Evaluation Results

Accuracy: 0.7567829457364341

Precision: 0.7210682492961603

Recall: 0.6075

F1 Score: 0.6594301221166894

Email ID	Event Name	Event Location	Date	Submission Deadline	Notification Deadline	Abstract Date	Published Date
0	International Conference on Green Computing, Technology and Innovation	Malaysia	March 4-6, 2013				
1	International Congress on Engineering and Information	Japan	May 6-8, 2020				
2	International Congress on Information and Communication Technology	United Kingdom	20 - 21 February 2020	30 November 2019			
3							
4							
5	International Conference on Signal Processing and Integrated Networks (	Noida	27- 28 February 2020				
6	International Conference on Digital Information and Communication Technology and its Applications	Thailand	May 06-08, 2014	April 06, 2014	26		
7				December		21,	
8	International Conference on Digital Information and Communication Technology and its Applications	Czech Republic	July 8-10, 2013	10, 2013		May	
9	International Conference on E-Learning and E-Technologies in Education	Kuala Lumpur	March 18-20, 2014	Feb. 18, 2014			
10	The Second International Conference on Informatics & Applications	Kuala Lumpur	July 8-10, 2013				
11	International Conference on Digital Information Networking, and Wireless Communications	Czech Republic	June 24-26, 2014	3 days	14, 2014		
12	International Conference on Digital Information Processing, E-Business and Cloud Computing	Dubai	October 23-25, 2013	.	18		
13	International Conference on Informatics Engineering & Information Science	Kuala Lumpur	Nov				
14							
15	International Symposium on Cyber Warfare, Cyber Defense, & Cyber Security						
16	International Symposium on SMART CITIES & SMART MOBILITY	Las Vegas	-	, 2019		21	
17		Las Vegas	July 27-30, 2015			May 31, 2015	
18	International Conference on Foundations	Las Vegas	July 27-30, 2015			March 31, 2015	
19	International Conference on Health	Las Vegas	July 27-30, 2015			May 31, 2015	
20	Internet Computing and Big Data	Las Vegas	July 27-30, 2015			March 31, 2015	
21	International Conference on Modeling	Las Vegas	July 27-30			March 31, 2015	

22	International Conference on Artificial Intelligence and Soft Computing						
23		Las Vegas	July 27-30, 2015			May 31, 2015	
24	Distributed Processing Techniques and Applications	Las Vegas	July 27-30, 2015			May 31, 2015	
25	International Conference on Image Processing	Las Vegas	July 27-30			April 15, 2015	
26	International Conference on Artificial Intelligence	Las Vegas	July 27-30			April 15, 2015	
27	World Congress on Computing and Information Technology		March 18-20, 2014	Feb. 18, 2014	08		
28			-04	03			
29	CSC'14	Las Vegas	July 21-24	Spring break		19.	
30	SAM'14	Las Vegas	July 21-24, 2014	Spring break		April 19, 2014	
31	SAM'14	Las Vegas	July 21-24, 2014	2014		April 19, 2014	
32	World Congress in Computer Science	Las Vegas	July 21-24,	2014		April 19,	
33	FECs'14	Las Vegas	July 21-24			March 15, 2014	
34							
35	International Conference on Internet Computing & IoT	Las Vegas	July 29 - August 1, 2019			June 20, 2019	
36	IPCV'14	Las Vegas	July 21-24, 2014			June 4, 2014	
37	'14	Las Vegas	July 21-24, 2014			June 4, 2014	
38	International Conference on Security and Management	Las Vegas	July 21-24, 2014			June 5, 2014	
39				November 1, 2014			
40	DA'	Las Vegas	July 27-30, 2015			March 31, 2015	
41	International Conference on Internet Computing	Las Vegas	July 27-30	March 31, 2015	24	31,	
42	ICAI'13	Las Vegas	July 22-25, 2013	2013		April 20, 2013	
43	BIOCOMP'13	Las Vegas	July 22-25, 2013	2013		April 20, 2013	
44	World Congress in Computer Science	Las Vegas	July 22-25			April 20, 2013	
45	Grid & Cloud Computing and Applications	Las Vegas	July 22-25, 2013			March 18, 2013	
46	International Conference on Scientific Computing	Las Vegas	July 22-25			March 18, 2013	
47	ICAI'13	Las Vegas	July 22-25, 2013	2013		April 6, 2013	
48	International Conference	Las Vegas	July 22-25,			March 18, 2013	
49	International Conference on	Las Vegas	July 22-25, 2013	2013		April 6, 2013	
50	World Congress in Computer Science	Las Vegas	July 22-25			April 6, 2013	
51	World Congress in Computer Science	Las Vegas	July 27-30, 2015			April 15, 2015	
52	BIOCOMP'15	Las Vegas	July 27-30	.		2015	
53	BIOCOMP'13	Las Vegas	July 22-25, 2013			June 2, 2013	
54	Software Engineering Research and Practice	Las Vegas	July 22-25, 2013			June 2, 2013	
55	World Congress in Computer Science	Las Vegas	July 21-24, 2014	1			
56				November 2014			
57							
58	International Conference on Digital Enterprise and Information Systems	Malaysia	March 4-6, 2013	January 3, 2013			
59	Ireland International Conference on Education	Dublin	April 20-22, 2015				

Default look with 128 testing emails results trained by our model

Email Dashboard

Paste text to process

Process Text

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Process Link

Model Evaluation Results

Accuracy: 0.7596899224806202

Precision: 0.7335329341317365

Recall: 0.716374269005848

F1 Score: 0.7248520710059171

Email ID	Event Name	Event Location	Date	Submission Deadline	Notification Deadline	Abstract Deadline	Publish Date	Decision Date
new_1765414124278	International Conference on Engineering Education and Innovation on December 06-08, 2025	Dubai, United Arab Emirates © 2025						
0			03-04 days	2019				
1	International Conference on Health In	Las Vegas	July 27-30, 2015	May 31, 2015				
2	International Conference on Internet Computing & IoT	Las Vegas	July 29 - August 1, 2019	June 20, 2019				

After clicking Process Link

Email Dashboard

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Process Link

Model Evaluation Results

Accuracy: 0.7596899224806202

Precision: 0.7335329341317365

Recall: 0.716374269005848

F1 Score: 0.7248520710059171

Email ID	Event Name	Event Location	Date	Submission Deadline	Notification Deadline	Abstract Deadline	Publish Date	Decision Date
new_1765414228370				March 20, 2014	March 25, 2014			
new_1765414124278	International Conference on Engineering Education and Innovation on December 06-08, 2025	Dubai, United Arab Emirates © 2025						
0			03-04 days	2019				
1	International Conference on Health In	Las Vegas	July 27-30, 2015	May 31, 2015				

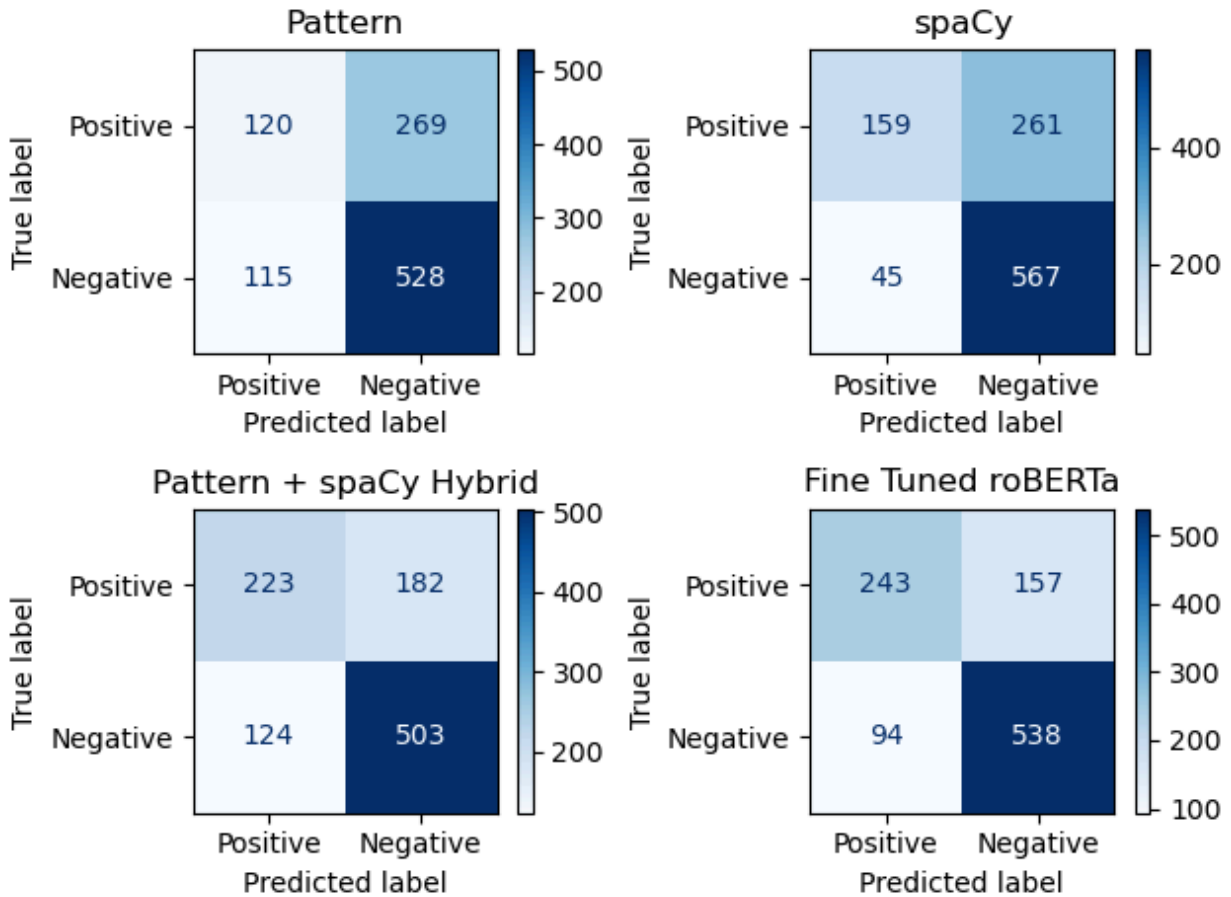
After clicking Process Text

Evaluation and Testing:

Our evaluation process started with a 20-80 test train split. This was done by taking random samples across all the emails. All models were trained using the training data and tested on using the test sample.

For each field TP, FP, FN, and TN metrics were generated allowing for the calculation of accuracy, precision, recall, and f1 scores. True Positives were calculated as the field being present in the human parsed and model recognised data, and having a fuzzy ratio of 20%.

Results from Evaluation:



Model	Accuracy	Precision	Recall	F1
Pattern	0.63	0.51	0.31	0.38
spaCy	0.70	0.78	0.38	0.51
Hybrid	0.70	0.64	0.55	0.59
roBERTa	0.76	0.72	0.61	0.66

Shortcomings:

In the beginning, we had difficulties trying to create a good model because most of us were new to machine learning and we haven't taken CSC 466 yet to learn the basics. And so a lot of the time, we had to do research to navigate our way through this project.

Our current model still isn't the best but we've tried our best to fine tune it. One consistent challenge across the NER model was high NULL output rates. When a field was not extracted, the evaluator treated this as a false negative, significantly lowering recall. The spaCy model reduced NULL outputs but still failed on non standard phrasing and underrepresented fields. Even with a hybrid fallback, NULLs for event names and deadlines kept accuracy modest, motivating the shift toward a transformer based model.

Human:

Actual Email:

Future Work:

One possible improvement is to support more input formats, such as PDF files and email attachments, since many conference announcements are not provided as plain text.

We also plan to improve extraction accuracy by using larger pretrained language models and fine-tuning them on domain-specific conference data.

In addition, we would like to add a human-in-the-loop interface that allows users to review and correct the extracted information, which could also help improve the model over time. Our goal is to improve overall performance, especially to bring the F1 score as close to 1 as possible.

Extra Images of Our Work:

Hand Tagging Email Extra Fields Sheet:

https://docs.google.com/spreadsheets/d/1MGVvIjV_35uNpYi0IsZXl3Jkj9oOCkph7I9jvt4KJiQ/edit?usp=sharing

A	B	C	D
ID	Abstract Date	Decision Date	Published Date
0			
1			
2			
3			
4			
5			
6	February 5, 2021		
7			
8	October 14, 2019		
9	October 14, 2019		
10			
11			
12	October 23		
13			
14			
15			
16			
17			
18			
19			
20			
21			
22			
23			
24			July 10, 2019
25			

Deployed Link: <https://482-conference-project.vercel.app/>